Project semester 1/2569 (20%)

Objective

The objective of this project is to learn and practice software project management by applying PMBOK concepts together with Generative AI. Each group will propose a software development project and use GenAI to assist in creating project management artifacts, evaluating alternatives, and supporting project decisions.

Team

1. Form a total of around 9 groups (3-5 students each). Use the same group as Requirements Engineering course.
2. Propose a software development project that you will manage. Use the project in the Requirement Engineering course.
3. Apply project management knowledge to deliver project management artifacts in your project report, submitted as three milestones.
4. Each member must actively participate in every milestone by creating prompts, analyzing GenAI outputs, revising artifacts, and contributing to shared project decision making.

GenAI Usage

Students may use any GenAI tool or model. The project will not be graded based on which GenAI tool is used or the quality of the initial GenAI output. Instead, grading emphasizes the process of using GenAI with human-in-the-loop which includes prompt design, evaluation, revision, and decision making. Students should assume GenAI is an assistant rather than a project manager.

Tasks

1. For each milestone, each member of the group will write prompts, evaluate the generated output, and revise prompts or perform manual edits.
2. See the descriptions and examples of the project artifacts in the lecture slides.
3. To reduce miscommunication among members, the group should establish shared understanding of the project context and project information. This creates a single source of truth to reduce inconsistency among generated outputs from different members.
4. For each milestone, each group member must contribute at least two prompts. The group should plan and distribute prompt engineering tasks to each member appropriately. That is, a simple artifact may require a single prompt, while a complex artifact may result from multiple prompts.

5. The recommended prompt frameworks are listed below. Clearly specify each part of the prompt text with the corresponding framework component, such as "**[character]** You are a senior project manager in a software company…"
   5.1. RTF for a quick request: Role (who the GenAI acts as, like a risk manager), Task (what specific work to do), and Format (how to display the result, like a table or bullet list). See PMI video https://youtu.be/kFajLF7W160?si=KTTpOK_0DTvm-Dcb
   5.2. CREATE for a more complex task: Character (persona, background, or professional mindset the GenAI must assume), Request (task or objective you need completed), Examples (samples to show the GenAI exactly what "good" looks like), Adjustments (specific project information, constraints, rules, or guidelines), Type of Output (structure of the final delivery, such as an executive summary or a markdown table), and Evaluation (evaluation criteria to fine-tune the results). See PMI video https://youtu.be/CuynTRSveLg?si=NH-x6zDzgLlK3TKK
6. Students should consider the following GenAI output evaluation criteria:
   6.1. Requirement Satisfaction: Does the output satisfy the requirement and information stated in the prompt? Is anything missing or incorrect?
   6.2. PMBOK Correctness: Is the output consistent with PMBOK concepts or terminology taught in class?
   6.3. Project Consistency: Is the output consistent with the project scope, assumptions, and previously developed artifacts?
7. Before milestone submission, all group members must form a Review Board to make a joint decision on final artifacts to be delivered for the milestone.
8. The final deliverables should be correct and complete but CONCISE. That is, you do not have to submit everything in the generated outputs. You may select necessary parts or summarize them where appropriate.

Submission

Milestone 1 (6%)

- Part 1 GenAI Log file (PDF)
- Part 2 Final Deliverables file (PDF)
  1. Review Board Decision Log
  2. Business case, containing background, business objective, current situation and problem, critical assumptions and constraints, analysis of problem resolution options and option recommendation, preliminary project requirements, budget estimate, schedule estimate, and potential risks.
  3. Financial analysis

4. Project charter, containing project title, project start date and finish date, budget information, project manager, project objectives, project success criteria, project approach, and stakeholder list.
5. Project organization chart

Milestone 2 (8%)

- Part 1 GenAI Log file (PDF)
- Part 2 Final Deliverables file (PDF)
  1. Review Board Decision Log
  2. Project scope statement, containing project summary, product characteristics and requirements, project deliverables including project management-related deliverables and product-related deliverables, what is excluded, and project success criteria.
  3. WBS and tasks
  4. Effort estimation for tasks
  5. Human and physical resource assignment to tasks and resource rates
  6. Cost estimation (labor costs, other costs, contingency reserve) that is aligned with resource assignment, resource rates, and efforts.
  7. Schedule based on WBS and tasks together with resource assignment to tasks (Gantt chart view and critical path)

Milestone 3 (6%)

- Part 1 GenAI Log file (PDF)
- Part 2 Final Deliverables file (PDF)
  1. Review Board Decision Log
  2. Quality management plan, including selected quality assurance activities, selected quality control tools and examples of how they are applied, and quality metrics.
  3. Risk register, including negative risks and positive risks, risk description, risk category, root cause, trigger, potential response, response strategy, risk owner, probability, impact, and status.
  4. Stakeholder communication analysis including stakeholder communications requirements, information to be communicated, who will produce it, methods or technologies for conveying the information, and frequency of communication.
- Part 3 Prompt Portfolio file (PDF) – Final key prompts used in all milestones

GenAI Log file

GenAI log file consists of all group members' prompt logs. Each member submits ONLY TWO prompt log records. Start each prompt log record on a new page. Each prompt log record has the following content.

1. Artifact	What you want AI to do or create
2. Member	Member contributing to prompt design
3. AI Tool	Tool/model used
4. Initial Prompt	Initial prompt used
5. Initial Output	Output of initial prompt or brief summary
6. Review	Review result of the initial output against evaluation criteria, and if not yet satisfied, identify weaknesses
7. Revision	How the initial prompt is revised or how the initial output is manually edited
8. Final Output	Final output and why it is acceptable

Note: You may perform review and revision several times. If so, just briefly summarize the results altogether and submit only one Final Output.

Review Board Decision Log

An example is shown below.

No.	Artifact	Member (prompt design)	Suggestion from team (in brief)	Decision (Accept / Revise / Reject)
1	Business case	Jane	-	Accept
2	Business case - preliminary project requirements	Alice	Add chatbot feature.	Revise
3	Budget estimates	Jane	Labor costs seem too low.	Revise
...				

Submission Date

All milestone submissions are to be placed in Course Dropbox on myCourseVille. In each milestone presentation, you should emphasize the project context and your experience of the iterative prompting and decision-making process. You may present only key prompting.

1. Thu 27 Aug: Submit group members, project name, and short project description in PDF
2. Thu 10 Sep: Submit milestone 1 in PDF
3. Thu 15 Oct: Submit milestone 2 in PDF
4. Sec 1 Tue 20 Oct, Sec 5 Sat 24 Oct: Presentation 1 in class for milestone 1+2.
5. Thu 12 Nov: Submit milestone 3 in PDF
6. Sec 1 Tue 17 Nov, Sec 5 Sat 21 Nov: Presentation 2 in class for milestone 3.

Scoring criteria

Scoring is determined with respect to

- PMBOK correctness
- Consistency between project artifacts
- Quality of critical reasoning and judgment during prompt engineering
- Quality of written and oral project presentation